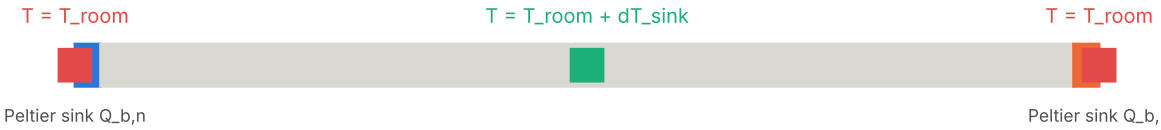
G5 Thermal boundary configurations (one model; studies switch the nodes on/off)

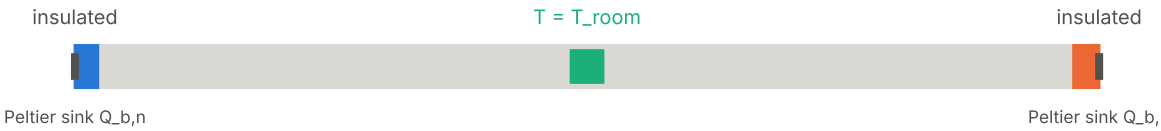
Config A (Studies 1 and 2): room at both ends, sink at the centre (Study 1: dT_sink = 0)

Output: heat taken from the room Q_room(dT_sink); dT_max = sink-room difference where Q_room = 0



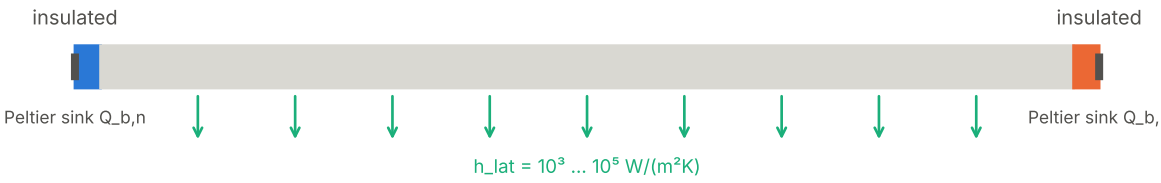
Config B (Study 3): insulated ends, sink at the centre

Output: how far the ends cool (or warm) below the sink: T_end - T_room



Config C (Study 4): insulated ends, lateral convective sink along the i-region

q_lat = h_lat (P/A)(T - T_out); 50 µm walls next to each junction insulated (w_wall)



In every configuration: volumetric heat source semi.Q_tot and boundary heat sources at both contacts. Semiconductor temperature fixed at T0 (one-way coupling).